# On $P_N$ and $SP_N$ Equations in Multidimensional Geometry

Dmitriy Y. Anistratov

Department of Nuclear Engineering
North Carolina State University

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Neutron Transport Theory
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# Outline


1. $P_N$ equations
   - Formulation of $P_N$ equations
   - Alternative form of $P_N$ equations
2. $SP_N$ Equations

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# Transport equation & spherical harmonics

$$\vec{\Omega}\cdot\vec{\nabla}\psi+\Sigma_t\psi(\vec{r},\vec{\Omega})=\frac{1}{4\pi}\left(\Sigma_s+\frac{1}{k}\nu_f\Sigma_f\right)\int_{4\pi}\psi(\vec{r},\vec{\Omega}')d\vec{\Omega}'$$

$$\psi(\vec{r},\vec{\Omega})=\sum_{\ell=0}^{\infty}\sum_{m=-\ell}^{\ell}\phi_{\ell,m}(\vec{r})Y_{\ell,m}(\vec{\Omega})$$

$$Y_{\ell,m}(\theta,\gamma)=\sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}P_\ell^m(\cos\theta)e^{im\gamma}$$

$$Y_{\ell,-m}(\theta,\gamma)=(-1)^m Y^*_{\ell,m}(\theta,\gamma)$$

$$P_\ell^m(\xi)=(-1)^m(1-\xi^2)^{\frac{m}{2}}\frac{d^mP_\ell(\xi)}{d\xi^m}\qquad m=0,1,2,\ldots$$

$$\int_0^{2\pi}\int_0^{\pi}Y_{\ell,m}(\theta,\gamma)Y^*_{\ell',m'}(\theta,\gamma)\sin\theta d\theta d\gamma=\delta_{\ell,\ell'}\delta_{m,m'}$$

$$\phi_{\ell,m}(\vec{r})=\int_{4\pi}Y^*_{\ell,m}(\vec{\Omega})\psi(\vec{r},\vec{\Omega})d\vec{\Omega}$$

# $P_N$ equations

$P_N$ approximation

$$\psi(\vec{r},\vec{\Omega}) = \sum_{\ell=0}^{N} \sum_{m=-\ell}^{\ell} \phi_{\ell,m}(\vec{r}) Y_{\ell,m}(\vec{\Omega})$$

$$\frac{1}{2}\frac{\partial}{\partial x}\left(-C_{\ell-1,m-1}\phi_{\ell-1,m-1}+D_{\ell+1,m-1}\phi_{\ell+1,m-1}+E_{\ell-1,m+1}\phi_{\ell-1,m+1}-F_{\ell+1,m+1}\phi_{\ell+1,m+1}\right)$$
$$+\frac{i}{2}\frac{\partial}{\partial y}\left(C_{\ell-1,m-1}\phi_{\ell-1,m-1}-D_{\ell+1,m-1}\phi_{\ell+1,m-1}+E_{\ell-1,m+1}\phi_{\ell-1,m+1}-F_{\ell+1,m+1}\phi_{\ell+1,m+1}\right)$$
$$+\frac{\partial}{\partial z}\left(A_{\ell-1,m}\phi_{\ell-1,m}+B_{\ell+1,m}\phi_{\ell+1,m}\right)+\Sigma_t\phi_{\ell,m} = \left(\Sigma_s+\frac{1}{k}\nu_f\Sigma_f\right)\phi_{0,0}\delta_{\ell,0}\delta_{m,0} \qquad i=\sqrt{-1}$$

$$0 \le \ell < +\infty\,, \quad -\ell \le m \le \ell$$

$$A_{\ell,m} = \sqrt{\frac{(\ell-m+1)(\ell+m+1)}{(2\ell+3)(2\ell+1)}}\,, \quad B_{\ell,m} = \sqrt{\frac{(\ell-m)(\ell+m)}{(2\ell+1)(2\ell-1)}}$$

$$C_{\ell,m} = \sqrt{\frac{(\ell+m+1)(\ell+m+2)}{(2\ell+3)(2\ell+1)}}\,, \quad D_{\ell,m} = \sqrt{\frac{(\ell-m)(\ell-m-1)}{(2\ell+1)(2\ell-1)}}$$

$$E_{\ell,m} = \sqrt{\frac{(\ell-m+1)(\ell-m+2)}{(2\ell+3)(2\ell+1)}}\,, \quad F_{\ell,m} = \sqrt{\frac{(\ell+m)(\ell+m-1)}{(2\ell+1)(2\ell-1)}}$$

# $P_1$ approximation

- $P_1$ approximation

$$\psi(\vec{r},\vec{\Omega}) = \sum_{\ell=0}^{1}\sum_{m=-\ell}^{\ell} \phi_{\ell,m}(\vec{r})\, Y_{\ell,m}(\vec{\Omega})$$

$$\psi(\vec{r},\vec{\Omega}) = \phi_{0,0}(\vec{r})\, Y_{0,0}(\vec{\Omega}) + \phi_{1,-1}(\vec{r})\, Y_{1,-1}(\vec{\Omega}) + \phi_{1,0}(\vec{r})\, Y_{1,0}(\vec{\Omega}) + \phi_{1,1}(\vec{r})\, Y_{1,1}(\vec{\Omega})$$

$$Y_{\ell,m}(\theta,\gamma) = \sqrt{\frac{2\ell+1}{4\pi}\frac{(\ell-m)!}{(\ell+m)!}}\, P_\ell^m(\cos\theta) e^{im\gamma}$$

$$Y_{0,0}(\theta,\gamma) = \sqrt{\frac{1}{4\pi}}\, P_0^0(\cos\theta)$$

$$Y_{1,-1}(\theta,\gamma) = -\sqrt{\frac{3}{8\pi}}\, P_1^1(\cos\theta) e^{-i\gamma}$$

$$Y_{1,0}(\theta,\gamma) = \sqrt{\frac{3}{4\pi}}\, P_1^0(\cos\theta)$$

$$Y_{1,1}(\theta,\gamma) = \sqrt{\frac{3}{8\pi}}\, P_1^1(\cos\theta) e^{i\gamma}$$

# Outline


1. $P_N$ equations
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2. $SP_N$ Equations

# Alternative Form of $P_N$ equations

- Cartesian geometry

$$\vec{\Omega} = (\Omega_1, \Omega_2, \Omega_3)\,, \quad \Omega_1 = \sin\theta\cos\gamma\,, \quad \Omega_2 = \sin\theta\sin\gamma\,, \quad \Omega_3 = \cos\theta$$

- Basis functions are linear combination of the spherical harmonic functions

$$f(\vec{\Omega}) = 1\,,$$

$$f_i(\vec{\Omega}) = \Omega_i\,, \quad i = 1,2,3$$

$$f_{ij}(\vec{\Omega}) = \frac{1}{2}(3\Omega_i\Omega_j - \delta_{ij})\,, \quad i,j = 1,2,3$$

$$f_{ijk}(\vec{\Omega}) = \frac{1}{2}\left(5\Omega_i\Omega_j\Omega_k - (\Omega_i\delta_{jk} + \Omega_j\delta_{ik} + \Omega_k\delta_{ij})\right), \quad i,j,k = 1,2,3$$

$$f_{ijk\ell}(\vec{\Omega}) = \frac{1}{8}\Bigg(35\Omega_i\Omega_j\Omega_k\Omega_\ell - 5\big(\Omega_i\Omega_j\delta_{k\ell} + \Omega_i\Omega_k\delta_{j\ell} + \Omega_i\Omega_\ell\delta_{jk} + \Omega_j\Omega_k\delta_{i\ell} + \Omega_j\Omega_\ell\delta_{ik} + \Omega_k\Omega_\ell\delta_{ij}\big) + \delta_{ij}\delta_{k\ell} + \delta_{ik}\delta_{j\ell} + \delta_{i\ell}\delta_{kj}\Bigg)\,, \quad i,j,k,\ell = 1,2,3$$

# Alternative form of $P_N$ equations

- Angular moments

$$\phi = \int f(\vec{\Omega})\psi d\Omega$$

$$\phi_i = \int f_i(\vec{\Omega})\psi d\Omega\,, \quad i = 1,2,3$$

$$\phi_{ij} = \int f_{ij}(\vec{\Omega})\psi d\Omega\,, \quad i,j = 1,2,3$$

$$\phi_{ijk} = \int f_{ijk}(\vec{\Omega})\psi d\Omega\,, \quad i,j,k = 1,2,3$$

$$\phi_{ijk\ell} = \int f_{ijk\ell}(\vec{\Omega})\psi d\Omega\,, \quad i,j,k,\ell = 1,2,3$$

# Alternative form of $P_N$ equations

- $P_3$ moment equations without closure in 3D

$$\sum_{m=1}^{3} \frac{\partial \phi_m}{\partial x_m} + \Sigma_{a,0}\phi = \frac{1}{k}\nu_f \Sigma_f \phi$$

$$\frac{1}{3}\frac{\partial \phi}{\partial x_i} + \frac{2}{3}\sum_{m=1}^{3} \frac{\partial \phi_{im}}{\partial x_m} + \Sigma_{a,1}\phi_i = 0\,, \quad i = 1,2,3$$

$$\frac{3}{5}\sum_{m=1}^{3} \frac{\partial \phi_{ijm}}{\partial x_m} + \frac{3}{10}\left(\frac{\partial \phi_i}{\partial x_j} + \frac{\partial \phi_j}{\partial x_i}\right) - \frac{1}{5}\delta_{ij}\sum_{m=1}^{3} \frac{\partial \phi_m}{\partial x_m} + \Sigma_{a,2}\phi_{ij} = 0\,, \quad i,j = 1,2,3$$

$$\frac{4}{7}\sum_{m=1}^{3} \frac{\partial \phi_{ijkm}}{\partial x_m} + \frac{5}{21}\left(\frac{\partial \phi_{ij}}{\partial x_k} + \frac{\partial \phi_{ik}}{\partial x_j} + \frac{\partial \phi_{jk}}{\partial x_i}\right)$$

$$- \frac{2}{21}\sum_{m=1}^{3} \frac{\partial}{\partial x_m}\left(\delta_{jk}\phi_{im} + \delta_{ik}\phi_{jm} + \delta_{ij}\phi_{km}\right) + \Sigma_{a,3}\phi_{ijk} = 0\,, \quad i,j,k = 1,2,3$$

# Alternative form of $P_3$ equations in 3D

- Standard closure: $\phi_{ijkm} = 0$

$$\sum_{m=1}^{3} \frac{\partial \phi_m}{\partial x_m} + \Sigma_{a,0}\phi = \frac{1}{k}\nu_f \Sigma_f \phi$$

$$\frac{1}{3}\frac{\partial \phi}{\partial x_i} + \frac{2}{3}\sum_{m=1}^{3} \frac{\partial \phi_{im}}{\partial x_m} + \Sigma_{a,1}\phi_i = 0\,, \quad i = 1,2,3$$

$$\frac{3}{5}\sum_{m=1}^{3} \frac{\partial \phi_{ijm}}{\partial x_m} + \frac{3}{10}\left(\frac{\partial \phi_i}{\partial x_j} + \frac{\partial \phi_j}{\partial x_i}\right) - \frac{1}{5}\delta_{ij}\sum_{m=1}^{3} \frac{\partial \phi_m}{\partial x_m} + \Sigma_{a,2}\phi_{ij} = 0\,, \quad i,j = 1,2,3$$

$$\frac{5}{21}\left(\frac{\partial \phi_{ij}}{\partial x_k} + \frac{\partial \phi_{ik}}{\partial x_j} + \frac{\partial \phi_{jk}}{\partial x_i}\right) - \frac{2}{21}\sum_{m=1}^{3} \frac{\partial}{\partial x_m}\left(\delta_{jk}\phi_{im} + \delta_{ik}\phi_{jm} + \delta_{ij}\phi_{km}\right) + \Sigma_{a,3}\phi_{ijk} = 0\,, \quad i,j,k = 1,2,3$$

# Formulation of the $SP_N$ equations

- $P_N$ equations in 1D slab geometry

$$\frac{n+1}{2n+1}\frac{d\phi_{n+1}}{dx}+\frac{n}{2n+1}\frac{d\phi_{n-1}}{dx}+\Sigma_{a,n}\phi_n=q\delta_{0,n}\,,\quad n=0,1,2,3...$$

$$\phi_n(x)=2\pi\int_{-1}^{1}P_n(\mu)\psi(x,\mu)d\mu$$

- Formulation of $SP_N$ equations
  1. $\frac{d}{dx}\rightarrow\frac{\partial}{\partial x_\alpha}$, $\alpha=1,2,3$
  2. $n$=odd: $\phi_n(x)\rightarrow\Phi_{n,\alpha}(\vec{x})$ and $\frac{d\phi_n}{dx}\rightarrow\vec{\nabla}\cdot\vec{\Phi}_n\equiv div\vec{\Phi}_n$
  3. $n$=even: $\phi_n(x)\rightarrow\Phi_n(\vec{x})$ and $\frac{d\phi_n}{dx}\rightarrow\frac{\partial\Phi_n}{\partial x_\alpha}$

# Formulation of $SP_N$ equations

- $n = 0, 2, 4, ...$

$$\sum_{\alpha=1}^{3} \left[ \frac{n+1}{2n+1} \frac{\partial \Phi_{n+1,\alpha}}{\partial x_\alpha} + \frac{n}{2n+1} \frac{\partial \Phi_{n-1,\alpha}}{\partial x_\alpha} \right] + \Sigma_{a,n} \Phi_n = q \delta_{0,n}$$

or

$$\left[ \frac{n+1}{2n+1} \vec{\nabla} \cdot \vec{\Phi}_{n+1} + \frac{n}{2n+1} \vec{\nabla} \cdot \vec{\Phi}_{n-1} \right] + \Sigma_{a,n} \Phi_n = q \delta_{0,n}$$

- $n = 1, 3, 5, ...$

$$\frac{n+1}{2n+1} \frac{\partial \Phi_{n+1}}{\partial x_\alpha} + \frac{n}{2n+1} \frac{\partial \Phi_{n-1}}{\partial x_\alpha} + \Sigma_{a,n} \Phi_{n,\alpha} = 0 \,, \quad \alpha = 1, 2, 3$$

or

$$\frac{n+1}{2n+1} \vec{\nabla} \Phi_{n+1} + \frac{n}{2n+1} \vec{\nabla} \Phi_{n-1} + \Sigma_{a,n} \vec{\Phi}_n = 0$$

- Interpretation of first two moments

$$\Phi_0 = \int_{4\pi} \psi d\Omega$$

$$\Phi_{1,\alpha} = \int_{4\pi} \Omega_\alpha \psi d\Omega$$

# Formulation of $SP_N$ equations

- $SP_N$ can be cast as coupled diffusion-like equations

$$\Phi_{n,\alpha} = -\frac{1}{\Sigma_{a,n}}\left[\frac{n+1}{2n+1}\frac{\partial \Phi_{n+1}}{\partial x_\alpha} + \frac{n}{2n+1}\frac{\partial \Phi_{n-1}}{\partial x_\alpha}\right] \qquad \alpha = 1,2,3$$

$$-\sum_{\alpha=1}^{3}\frac{\partial}{\partial x_\alpha}\left(\frac{n+1}{(2n+1)\Sigma_{a,n+1}}\frac{\partial}{\partial x_\alpha}\left[\frac{n+2}{2n+3}\Phi_{n+2} + \frac{n+1}{2n+3}\Phi_n\right]\right.$$
$$\left. + \frac{n}{(2n+1)\Sigma_{a,n-1}}\frac{\partial}{\partial x_\alpha}\left[\frac{n}{2n-1}\Phi_n + \frac{n-1}{2n-1}\Phi_{n-2}\right]\right) + \Sigma_{a,n}\Phi_n = q\delta_{0,n}$$

# $SP_3$ equations in 3D

$$\sum_{\alpha=1}^{3} \frac{\partial \Phi_{1,\alpha}}{\partial x_\alpha} + \Sigma_{a,0}\Phi_0 = q$$

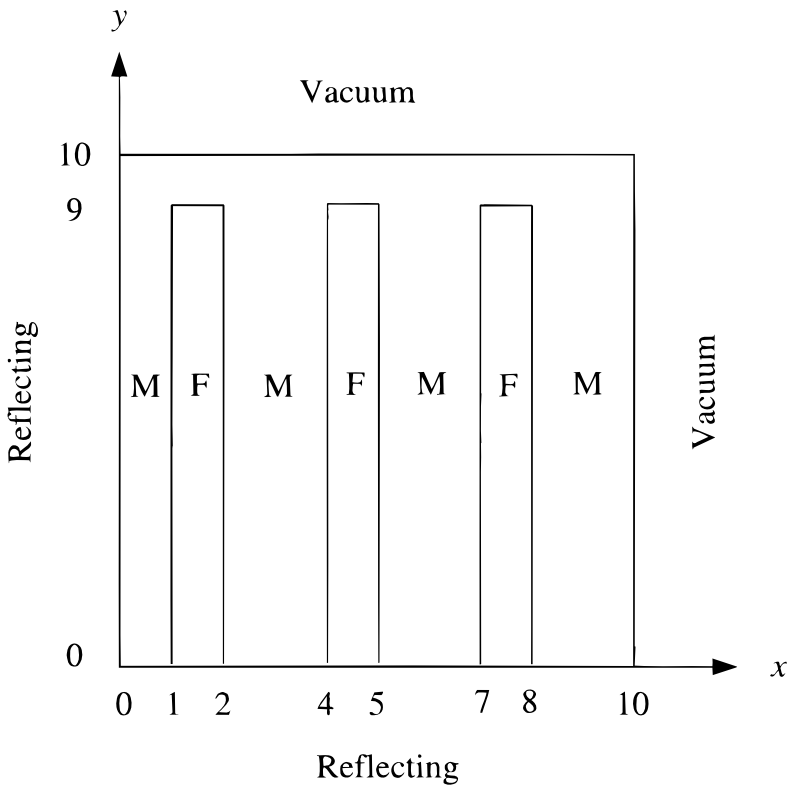
$$\frac{2}{3}\frac{\partial \Phi_2}{\partial x_\alpha} + \frac{1}{3}\frac{\partial \Phi_0}{\partial x_\alpha} + \Sigma_{a,1}\Phi_{1,\alpha} = 0\,, \quad \alpha = 1,2,3$$

$$\frac{3}{5}\sum_{\alpha=1}^{3} \frac{\partial \Phi_{3,\alpha}}{\partial x_\alpha} + \frac{2}{5}\sum_{\alpha=1}^{3} \frac{\partial \Phi_{2,\alpha}}{\partial x_\alpha} + \Sigma_{a,2}\Phi_2 = 0$$

$$\frac{3}{7}\frac{\partial \Phi_2}{\partial x_\alpha} + \Sigma_{a,3}\Phi_{3,\alpha} = 0\,, \quad \alpha = 1,2,3$$

# Numerical results $SP_3$ method

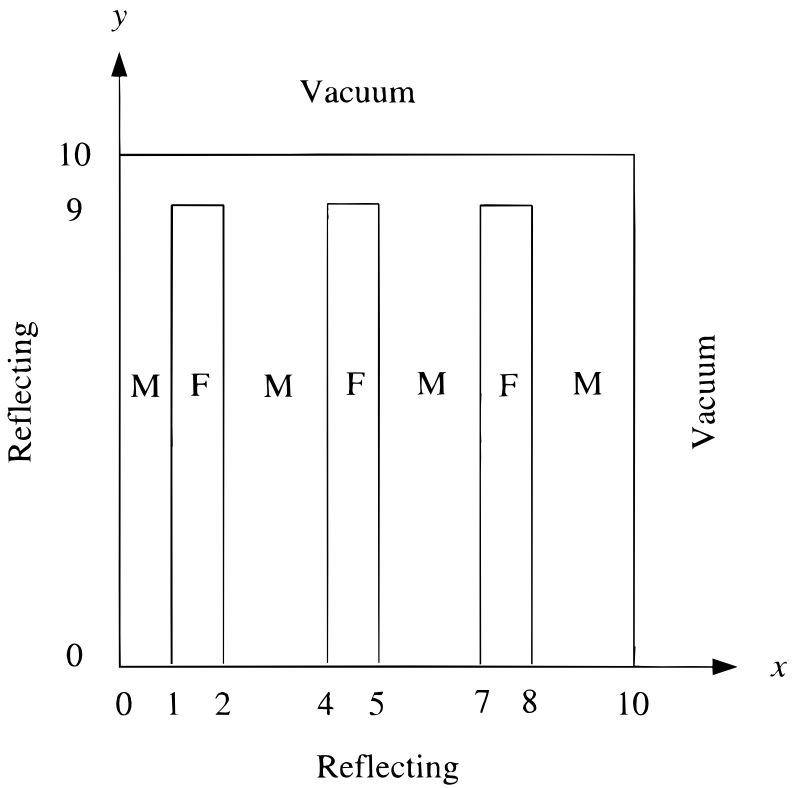
- Brantley & Larsen, The Simplified $P_3$ Approximation, NS&E. v. 134, 2000.




One-group k-eigenvalue problem

# Numerical results $SP_3$ method

- Brantley & Larsen, The Simplified $P_3$ Approximation, NS&E. v. 134, 2000.




One-group k-eigenvalue problem

Computed Eigenvalues for One-Group Eigenvalue Problem

| Method | $k$ | % Relative Error |
|---|---|---|
| $P_1$ | 0.776534 | −3.672 |
| $SP_3$ | 0.798617 | −0.932 |
| $S_{16}$ | 0.806132 | Reference |

# Numerical results $SP_3$ method

- Brantley & Larsen, The Simplified $P_3$ Approximation, NS&E, v. 134, 2000.

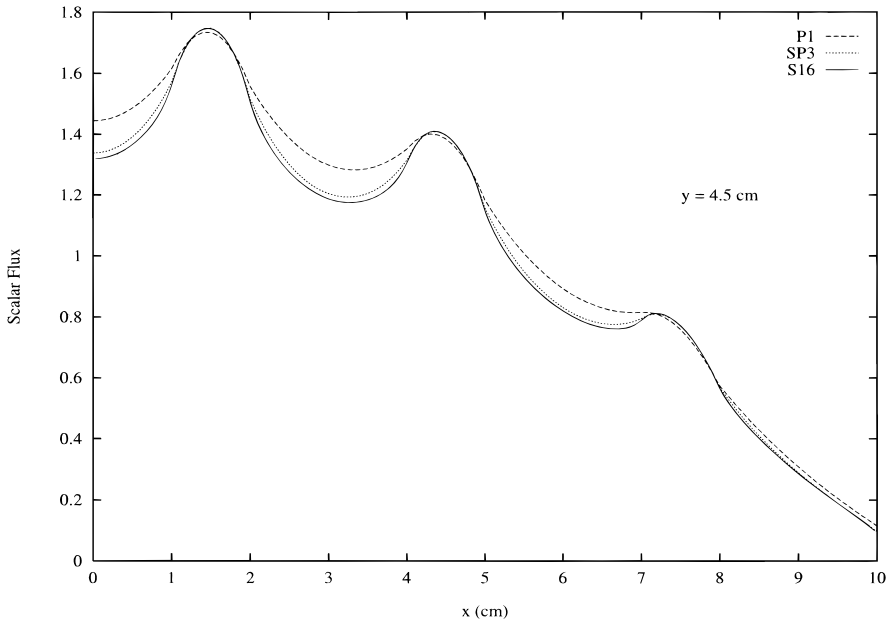



Fig. 4. Scalar flux along $y = 4.5$ cm for one-group eigenvalue problem.

# Numerical results $SP_3$ method

- Brantley & Larsen, The Simplified $P_3$ Approximation, NS&E, v. 134, 2000.

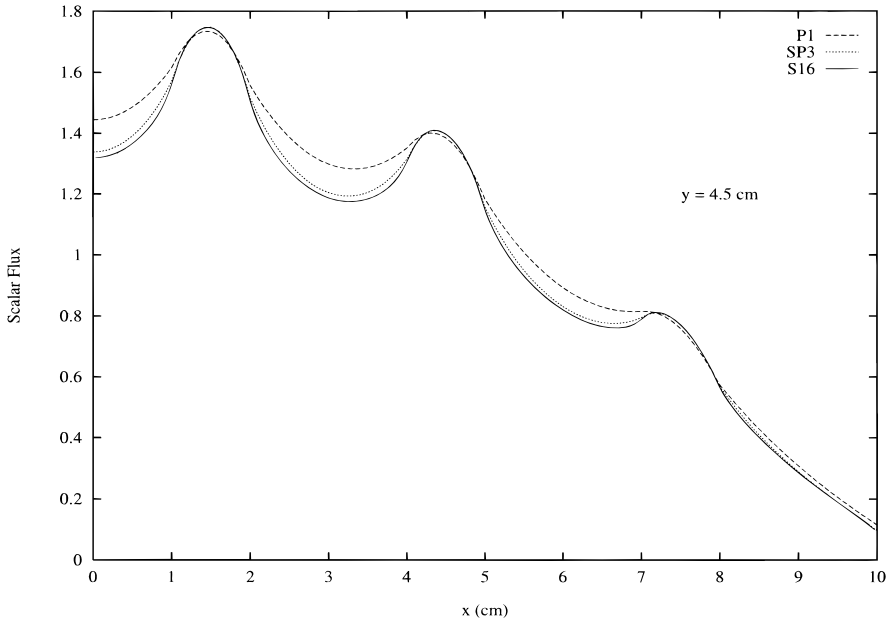



Fig. 4. Scalar flux along $y = 4.5$ cm for one-group eigenvalue problem.

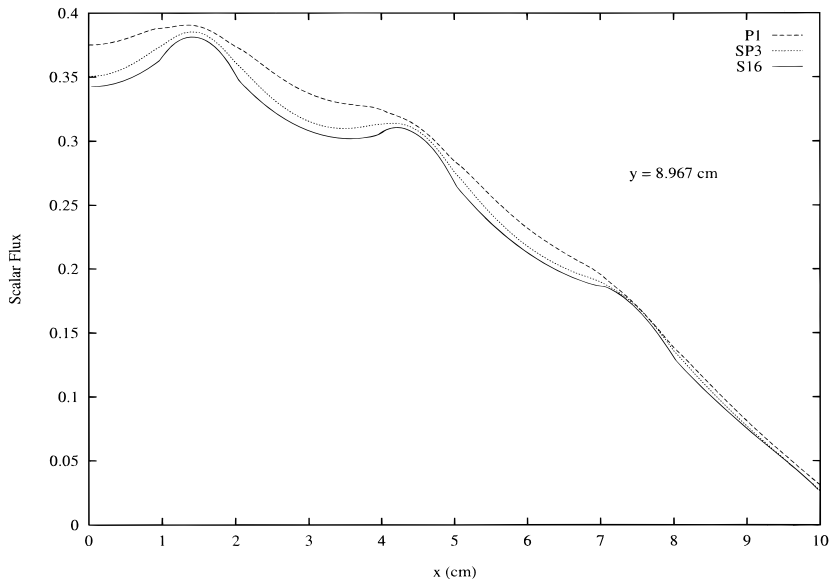



Fig. 5. Scalar flux along $y = 8.967$ cm for one-group eigenvalue problem.

# 23. Projection Operator Approach

$$T\psi = q \qquad T = L - S$$

Step 1. Define a projection operator $\mathcal{P}$ for the equation

$$\mathcal{P}T\psi = \mathcal{P}q$$

$$\Rightarrow \quad B\psi = Q \qquad B \triangleq \mathcal{P}T \quad Q = \mathcal{P}q$$

Step 2. Define a projection operator $\mathcal{F}$ for the solution of the projected equation

$$\vec{\varphi} = \mathcal{F}\psi$$

$$\Rightarrow \quad \tilde{B}\vec{\varphi} = Q$$

Step 3. Define a restriction (prolongation) operator $\mathcal{R}$ for the solution itself

$$\vec{\psi} = \mathcal{R}\vec{\varphi} \quad \Rightarrow \psi \approx \vec{\psi}$$

Example $P_N$

$$\mathcal{P} = \left\{ 2\pi \int_{-1}^{1} \cdot P_\ell(\mu) d\mu \right\}_{\ell=0}^{N} \equiv \mathcal{P}_N$$

$$\mathcal{F} = \left\{ 2\pi \int_{-1}^{1} \cdot P_\ell(\mu) d\mu \right\}_{\ell=0}^{N} = \mathcal{F}_N = \mathcal{P}_N$$

$$\vec{\varphi} = \left\{ \phi_0, \phi_1, \phi_2, \cdots \phi_N \right\}$$

$$\mathcal{R}: \quad \mathcal{P}\vec{\varphi} = \sum_{\ell=0}^{N} \frac{2\ell+1}{4\pi} \phi_\ell(x) P_\ell(\mu)$$

DP$_N$

$$\mathcal{P} = \left\{ 2\pi \int_{-1}^{1} \cdot P_\ell^{\pm}(\mu) d\mu \right\}_{\ell=0}^{N}$$

$$\mathcal{F} = \left\{ 2\pi \int_{-1}^{1} \cdot P_\ell^{\pm}(\mu) d\mu \right\}_{\ell=0}^{N}$$

$$\vec{\varphi} = \left\{ \phi_0^{\pm}, \phi_1^{\pm}, \cdots \phi_N^{\pm} \right\}$$

$$\mathcal{R}: \quad \mathcal{R}\vec{\varphi} = \sum_{\ell=0}^{N} \frac{2\ell+1}{4\pi} \left( \phi_\ell^{+}(x) P_\ell^{+}(\mu) + \phi_\ell^{-}(x) P_\ell^{-}(\mu) \right)$$

# 24. Quasidiffusion Method for Solving the Transport Equation

$$\mu \frac{\partial \psi}{\partial x}(x,\mu) + \Sigma_t(x)\psi(x,\mu) = \frac{1}{2}\Sigma_s(x) \int_{-1}^{1} \psi(x,\mu')d\mu' + \frac{1}{2}q(x)$$

$$\psi|_{x=0} = \psi_{in}^{+}(\mu) \quad \mu > 0 \qquad \psi|_{x=X} = \psi_{in}^{-}(\mu) \quad \mu < 0$$

$\int_{-1}^{1} \cdot d\mu$ and $\int_{-1}^{1} \cdot \mu\, d\mu$ ← projection operators for the equation

$$\mathcal{P} = \left\{ \int_{-1}^{1} \cdot \mu^n d\mu \right\}_{n=0,1}$$

$$\frac{d}{dx} J(x) + (\Sigma_t(x) - \Sigma_s(x))\phi(x) = q(x)$$

$$\frac{d}{dx} \int_{-1}^{1} \mu^2 \psi(x,\mu) d\mu + \Sigma_t(x) J(x) = 0$$

Note $\phi = \phi_0 \qquad J = \phi_1$

- QD method defines exact closure to the moment equations

$$\int_{-1}^{1} \mu^2 \psi(x,\mu) d\mu = E(x)\phi(x)$$

$$E(x) = \frac{\int_{-1}^{1} \mu^2 \psi(x,\mu) d\mu}{\int_{-1}^{1} \psi(x,\mu) d\mu}$$ ← Eddington factor / QD factor

$\Rightarrow$ Low-order QD equations
(aka Variable Eddington Factor equations)

$$\begin{cases} \dfrac{dJ}{dx} + (\Sigma_t - \Sigma_s)\phi = q \\ \dfrac{d}{dx}(E\phi) + \Sigma_t J = 0 \end{cases}$$ Low-order QD equations

$$\mu \frac{\partial \psi}{\partial x} + \Sigma_t \psi = \frac{1}{2}(\Sigma_s \phi + q)$$ high-order transport equation